

Single Shot 2AS: Design, Architecture, and Implementation of an Edge-to-Cloud Multi-Camera Automated Attendance System

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Executive Abstract

Biometric attendance management in educational institutions has long oscillated between intrusive, slow hardware turnstiles (e.g., optical fingerprint scanners, RFID readers) and expensive proprietary surveillance software. This paper details the complete technical architecture and production implementation of **Single Shot 2AS** (Single Shot Automated Attendance System), a production-engineered, cost-neutral, edge-to-cloud attendance management platform. Single Shot 2AS captures and authenticates an entire cohort of 35 to 60 students simultaneously from a single high-resolution classroom frame.

The architecture bridges physical-layer computer vision challenges with cloud synchronization through six synchronized subsystems:

1. **Edge Node Ingestion:** A centralized Master Node (Raspberry Pi 4 or Linux PC) orchestrating multi-camera RTSP/LAN streams with thread-safe hardware access and exposure warm-up.
2. **Quality & Standards-Compliant Vision Pipeline:** Image validation against international biometric standards (IEC 62676-4 DORI $\geq 80 \times 80$ px, ISO/IEC 19794-5 IPD ≥ 30 px), adaptive resolution preservation preventing back-row face annihilation, and multi-scale detection (OpenCV YuNet FPN with dlib HOG/CNN fallbacks) linked to 128-dimensional metric embeddings.
3. **Zero-Cost Relational Cloud Ledger:** A Google Drive and Google Sheets backend operating entirely within Google Cloud's free 5GB tier, utilizing Google OAuth 2.0 PKCE, dynamic column scaling, and atomic batch updates ($O(1)$ API calls).
4. **Voice Command & Acoustic Intimation:** An offline, on-premise acoustic pipeline utilizing Vosk (Kaldi-based) speech recognition with debounce gating and pyttsx3/espeak speech synthesis announcing gender-segregated roll-call tallies.
5. **Security Hardening:** Bytecode-restricted deserialization ([RestrictedUnpickler](#) blocking Remote Code Execution), state-verified OAuth CSRF prevention, API query sanitization, and role-gated MJPEG streams.
6. **Unified Web Management:** A Flask single-page application (SPA) featuring an automated four-stage onboarding wizard, multi-angle (Front/Left/Right) enrollment, and real-time dashboarding.

The entire codebase has been verified under automated test suites consisting of 103 unit and integration tests with zero regressions.

1. Introduction and Architectural Motivation

1.1 The Classroom Attendance Bottleneck

Roll-call processes in higher education institutions routinely consume between 7% and 12% of total instructional time. While automated fingerprint scanners and smart cards offer marginal improvements, they introduce mechanical bottlenecks: students must queue sequentially at a single terminal, causing hallway congestion and physical device wear.

Conversely, attempting facial recognition across an entire classroom in a **single wide-angle shot** presents three severe computer vision and systems challenges:

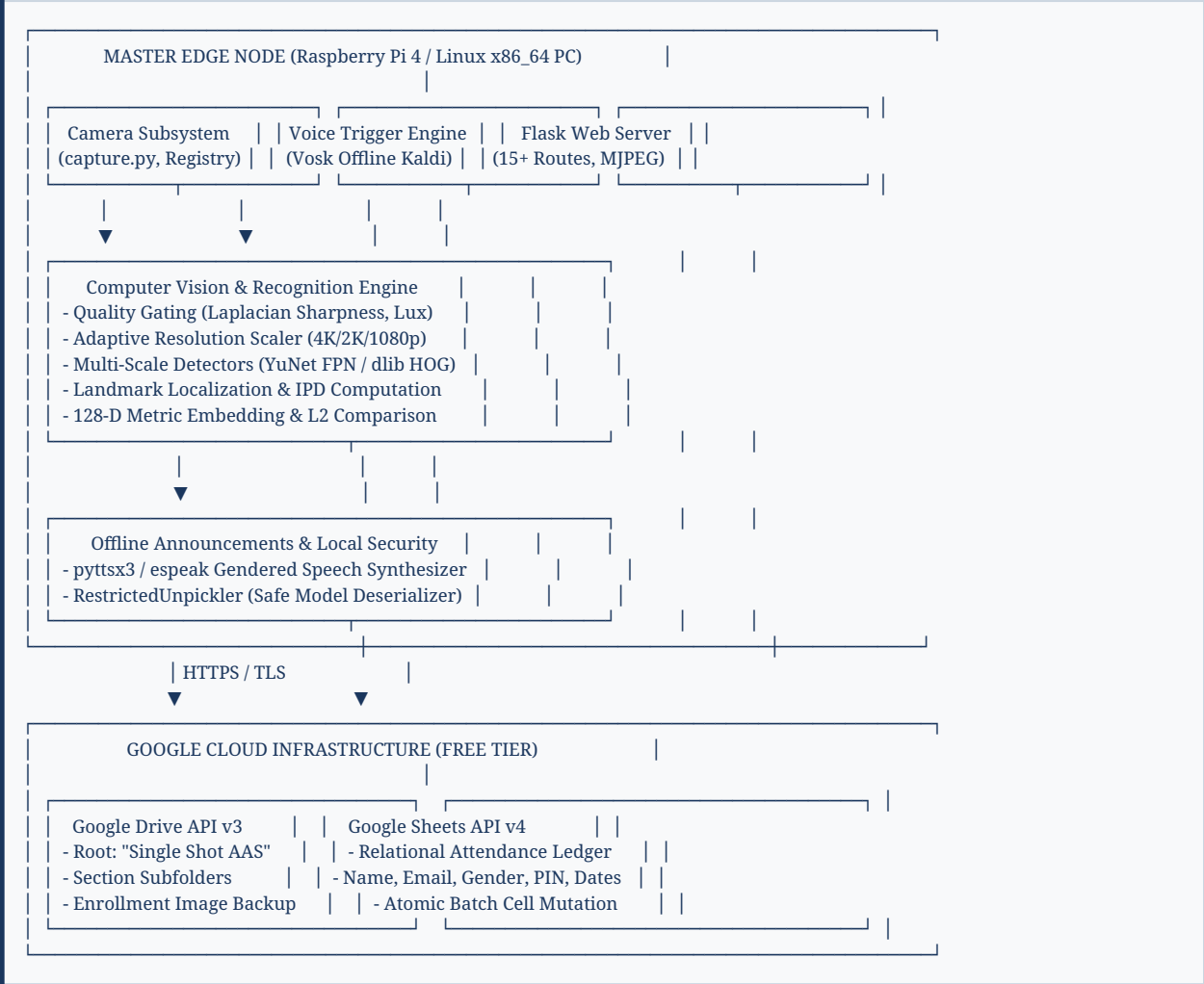
- 7. **The Perspective Resolution Drop-off:** In a 40-to-60 student classroom spanning 8 to 12 meters in depth, students in the front row occupy upwards of 300×300 pixels per face, whereas students in the rear row occupy as little as 30×30 pixels on standard 1080p sensors. Downscaling images indiscriminately destroys landmark features for rear students.
- 8. **Operational Infrastructure Costs:** Cloud-hosted biometric computer vision APIs (e.g., AWS Rekognition, Azure Face) charge per-face detection fees. In a college with 3,000 students across 6 periods daily, recurring cloud vision costs exceed \$4,000 annually.
- 9. **Data Sovereignty and Network Fragility:** Rural or campus-edge environments experience intermittent internet outages. Cloud-dependent biometric systems fail entirely when WAN connectivity drops.

1.2 System Goals of Single Shot 2AS

Single Shot 2AS is designed from first principles to resolve these constraints:

- **Single-Shot Concurrency:** Process an entire classroom cohort (35–60 individuals) in a single snapshot captured via ceiling-mounted RTSP IP cameras or faculty-triggered webcams.
- **\$0 Operational Cost:** Utilize Google Cloud Platform's permanent free tier (5 GB Google Drive storage and Google Sheets API v4) as the distributed relational database and document repository.
- **Local Edge Autonomy:** Keep all computer vision inference, voice recognition, and acoustic announcements 100% offline on an on-premise Master Node (Raspberry Pi 4 Model B or x86 Linux PC). Internet is required strictly for periodic ledger synchronization.
- **Standards-Compliant Reliability:** Explicitly reject degraded face crops failing international surveillance and biometric passport standards (IEC 62676-4 and ISO/IEC 19794-5) to prevent false-positive misidentifications.





2. System Prerequisites, Installation Lifecycle & Bootstrap

Single Shot 2AS is structured for deterministic, automated deployment across Unix-like platforms (Ubuntu, Debian, Raspberry Pi OS 64-bit) through a shell bootstrap sequence (`setup.sh`) and container virtualization.

2.1 Hardware Specifications

Subsystem	Minimum Specification	Recommended Production Specification
Master Node	Dual-core x86_64 or ARM Cortex-A72, 2 GB RAM	Raspberry Pi 4 Model B (4 GB / 8 GB RAM) or Intel Core i5/i7 Workstation
Storage	8 GB available flash/disk storage	32 GB Class 10 High-Endurance MicroSD or M.2 NVMe SSD

Subsystem	Minimum Specification	Recommended Production Specification
Camera Ingestion	1080p (1920×1080) USB UVC Webcam	4K (3840×2160) or 5MP H.264/H.265 RTSP Ceiling-Mounted IP Camera
Acoustic Input	Integrated USB microphone	USB Boundary / Conference Microphone (Omnidirectional, 16 kHz sampling)
Acoustic Output	Onboard 3.5mm headphone jack	3W–5W Powered Active Classroom Mini-Speaker
Local Networking	100 Mbps Fast Ethernet	Gigabit Ethernet (Cat6) connecting Master Node to CCTV Switch/VLAN

2.2 Operating System Dependencies & Toolchain

The underlying face recognition algorithms rely on **dlib**, which requires C++11 compilation tools, BLAS/LAPACK linear algebra acceleration, and PortAudio drivers for microphone capturing.

```
# Debian / Ubuntu / Raspberry Pi OS Package Manifest
sudo apt-get update && sudo apt-get install -y \
  build-essential \
  cmake \
  libboost-all-dev \
  libopenblas-dev \
  liblapack-dev \
  libx11-dev \
  portaudio19-dev \
  git \
  wget \
  unzip \
  espeak
```

2.3 Python Runtime & Virtual Environment

Single Shot 2AS requires **Python 3.10.x** to guarantee package binary compatibility with dlib wheels and Google client libraries.

```
# 1. Environment initialization via pyenv
pyenv install 3.10.20
pyenv local 3.10.20

# 2. Isolated Virtual Environment creation
python3 -m venv .venv
source .venv/bin/activate
```

```
# 3. Core dependencies installation
pip install --upgrade pip setuptools wheel
pip install -r requirements.txt
pip install -r requirements-dev.txt
```

2.4 Offline Acoustic Model Deployment

Voice trigger capabilities depend on the **Vosk** offline Kaldi-based automatic speech recognition (ASR) engine. The setup script downloads the English acoustic model (**vosk-model-small-en-us-0.15** , 40 MB) and mounts it at **<repo-root>/vosk-model-small-en-us/** :

```
wget -q https://alphacephei.com/vosk/models/vosk-model-small-en-us-0.15.zip -O vosk_model.zip
unzip -q vosk_model.zip
mv vosk-model-small-en-us-0.15 vosk-model-small-en-us
rm vosk_model.zip
```

2.5 Central Configuration Engine (`aas.core.config`)

All parameters, paths, and environment variables are centralized in **src/aas/core/config.py** . The configuration enforces strict absolute path calculation anchored to **ROOT_DIR** , preventing working-directory drift:

```
# Directory Anchor Hierarchy
CORE_DIR = os.path.dirname(os.path.abspath(__file__)) # src/aas/core/
SRC_DIR = os.path.dirname(os.path.dirname(CORE_DIR)) # src/
ROOT_DIR = os.path.dirname(SRC_DIR) # Single Shot AAS/

# Data and Runtime Layout
DATA_DIR = os.path.join(ROOT_DIR, "data")
PHOTO_FOLDER = os.path.join(DATA_DIR, "known_face_photos")
ENCODINGS_FOLDER = os.path.join(DATA_DIR, "known_face_encodings")
CAPTURED_FOLDER = os.path.join(DATA_DIR, "captured")
CONFIG_DIR = os.path.join(ROOT_DIR, "config")
CAMERAS_JSON_PATH = os.path.join(CONFIG_DIR, "cameras.json")
INSTITUTION_JSON_PATH = os.path.join(CONFIG_DIR, "institution.json")
```

Key environment configurations defined in **.env** :

```
# Google OAuth 2.0 Credentials
GOOGLE_CLIENT_ID=your-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=your-client-secret
OAUTH_REDIRECT_URI=http://localhost:5000/auth/google/callback

# Recognition & Vision Standards Configuration
RECOGNITION_TOLERANCE=0.55
RECOGNITION_SCALE=0.5
```

```

AUTO_SCALE_DETECTION=true
MIN_FACE_SIZE=80
MIN_IPD_PIXELS=30
BLUR_THRESHOLD=40.0
MIN_IMAGE_WIDTH=1280
RECOMMENDED_IMAGE_WIDTH=1920
FACE_DETECTION_MODEL=hog

# Voice and Audio Parameters
VOICE_TRIGGER_PHRASE=take attendance
MAX_IN_TIME=16:00:00

```

2.6 System Daemonization & Docker Virtualization

For production headless deployment on an edge device (e.g., Raspberry Pi 4 mounted in a classroom control cabinet), **attendance.service** provisions an auto-starting, auto-restarting systemd unit:

```

[Unit]
Description=Single Shot AAS - AI Attendance System Web & Edge Service
After=network.target network-online.target
Wants=network-online.target

[Service]
Type=simple
User=giri
WorkingDirectory=/home/giri/Projects/Single Shot 2A/Single Shot AAS
Environment="PYTHONPATH=src"
Environment="PATH=/home/giri/Projects/Single Shot 2A/Single Shot AAS/.venv/bin:/usr/local/bin:/usr/bin"
ExecStart=/home/giri/Projects/Single Shot 2A/Single Shot AAS/.venv/bin/python main.py --web
Restart=always
RestartSec=5
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target

```

Alternatively, a multi-stage **Dockerfile** handles compilation of dlib, OpenCV, and Vosk within a clean Debian Slim image, preventing build artifact pollution on edge storage.

3. End-to-End Operational Lifecycle & Flow Architecture

The operational workflow conforms to the sequential state machine detailed below, implementing user routing, automated setup provisioning, and live operation.

```
stateDiagram-v2
```

```

[*] --> SplashScreen: GET /splash
SplashScreen --> LoginPage: Redirect after 2000ms
LoginPage --> GoogleOAuth: User clicks "Login with Google"
GoogleOAuth --> OAuthCallback: Consent Granted & Token Received
OAuthCallback --> Verification: check_drive_setup_exists()

state Verification <<choice>>
Verification --> SetupWizard: If Root Folder Missing (New User)
Verification --> Dashboard: If Root Folder Exists (Returning User)

state SetupWizard {
  [*] --> Step1_Institution: Metadata (Org Name, City)
  Step1_Institution --> Step2_Sections: Define Sections & Spreadsheets
  Step2_Sections --> Step3_Cameras: Map RTSP Cameras & Test Frames
  Step3_Cameras --> Step4_Complete: Commit to Drive & Config
}

SetupWizard --> Dashboard: Setup Finalized

state Dashboard {
  [*] --> Idle
  Idle --> Enrollment: Web Form / 3-Angle Capture
  Idle --> TriggerShot: Web Click / Voice Trigger / CLI
  TriggerShot --> CaptureFrame: RTSP / Webcam Mutex Capture
  CaptureFrame --> VisionPipeline: Ingest Frame
  VisionPipeline --> CloudSync: Batch Update Google Sheet
  CloudSync --> LocalTTS: Speak Gendered Intimation
  LocalTTS --> EmailAlerts: Dispatch Confirmation Emails
}

```

3.1 Phase 1: Splash and Single-Sign-On Entry

- **Splash Screen** (`/splash`): Serves a branded initial loading view, querying `/api/auth/status` asynchronously.
- **Authentication Gate** (`/auth/google`): Bypasses local passwords. The system uses **Google OAuth 2.0 PKCE** with Google Drive and Google Sheets authorization scopes:
- <https://www.googleapis.com/auth/spreadsheets>
- <https://www.googleapis.com/auth/drive>
- <https://www.googleapis.com/auth/userinfo.profile>
- <https://www.googleapis.com/auth/userinfo.email>
- **CSRF Protection:** Every authorization request generates a cryptographically randomized 32-byte hexadecimal state string stored in the server session. The redirect handler (`/auth/google/callback`) verifies `request.args.get('state') == session['oauth_state']`, aborting on mismatch.

3.2 Phase 2: Autonomous State Verification

Upon successful OAuth token acquisition, the system executes `drive_manager.check_drive_setup_exists(service)`.

- It queries the user's Google Drive for a folder named exactly **"Single Shot AAS"**.

- **Branch A (New User):** No folder detected. The user is redirected to the Setup Wizard (`/setup`).
- **Branch B (Returning User):** Folder detected. Active spreadsheets and configurations are re-indexed, and the user is routed directly to the Dashboard (`/dashboard`).

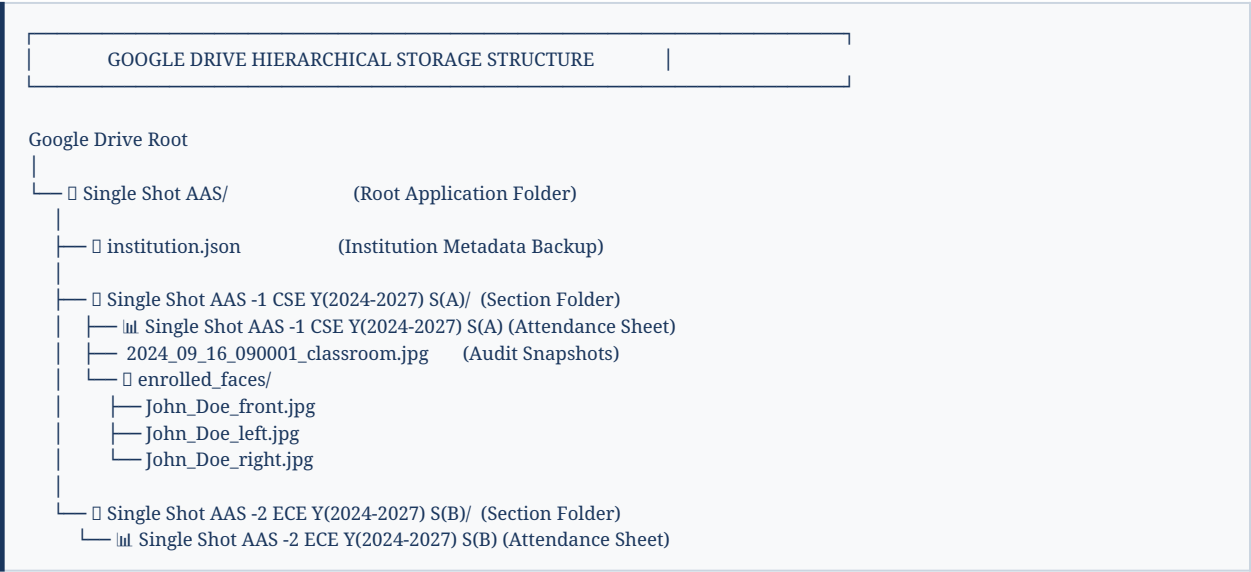
3.3 Phase 3: The Setup Wizard

The Setup Wizard systematically builds the institution's cloud data structure:

10. **Institution Registration (`~/setup/step1``):** Captures organization name, academic year, campus city, and administrator email, generating `config/institution.json` .
11. **Section & Ledger Provisioning (`~/setup/step2``):** Generates standardized sections. For each section, the Drive Manager provisions a designated Drive subfolder and initializes a Google Spreadsheet formatted with the strict naming specification.
12. **Camera Ingestion Mapping (`~/setup/step3``):** Registers RTSP network cameras or USB webcam devices to their respective section sheets. The backend performs active ping testing via `cv2.VideoCapture` to validate frame decoding before committing to `config/cameras.json` .

4. Cloud Ledger & Database Management Architecture

Single Shot 2AS replaces expensive database engines (e.g., PostgreSQL, MongoDB, Cloud SQL) with a hierarchical Google Drive and Google Sheets storage infrastructure, eliminating operational database administration while staying within the free tier.



Where:

- *X* (Prefix): Serial index of spreadsheet creation within the organization (*1*, *2*, *3*, ...).
- *xxxx*: Department or academic branch identifier (e.g., *CSE* , *ECE* , *MECH* , *CIVIL*).
- *Y(XXXX-XXXX)*: Cohort duration representing admission and graduation years (e.g., *Y(2024-2027)*).
- *S(X)*: Section identifier (e.g., *S(A)* , *S(B)* , *S(1)* , *S(2)*).

Implementation (`drive_manager.format_sheet_name`):

```
def format_sheet_name(serial: int, section_code: str, year_start: int,
                      year_end: int, section: str) -> str:
    return f"Single Shot AAS -{serial} {section_code} Y({year_start}-{year_end}) S({section})"
```

4.2 Relational Worksheet Schema

Within each spreadsheet, the primary worksheet (*Attendance*) functions as an append-only relational ledger:

Column	Header	Type	Constraints / Description
Col 1	<i>Name</i>	String	Unique student identifier; matches facial encoding ID.
Col 2	<i>Email</i>	String	Student/Guardian email for notification dispatches.
Col 3	<i>Gender</i>	Char(1)	Demographic marker: ' <i>M</i> ' (Male), ' <i>F</i> ' (Female), or ' <i>O</i> ' (Other).
Col 4	<i>PIN</i>	String	Unique numeric enrollment security PIN.
Col 5+	Date Columns	String	Dynamic date stamps formatted as locale-safe <i>M/D/YYYY</i> (e.g., <i>9/16/2026</i>). Values: <i>present</i> , <i>absent</i> , <i>late</i> .

4.3 High-Throughput Batch Mutation & Concurrency

A known failure mode in naive Google Sheets implementations is hitting Google's rate limit (**60 write requests per minute per user**). Calling `sheet.update_cell()` sequentially for 60 students consumes 60 API requests, causing immediate *429 Too Many Requests* exceptions.

Single Shot 2AS solves this through **Atomic Batch Mutation** (`SpreadsheetManager.write_batch_to_sheet`):

13. **Header Inspection:** Retrieves Row 1 in a single read call (`ws.row_values(1)`). If today's date column is absent, it is appended dynamically (`_ensure_date_column`).

14. **In-Memory Range Assembly:** Maps all present students to their corresponding row indices using exact string matching on Column 1 (`^Name$`).
15. **Atomic Write Payload:** Compiles an update list using `gspread.utils.rowcol_to_a1` :

$$\text{text}\{Cell\ Range\} = \text{text}\{ColLetter\} + \text{text}\{RowNum\}$$

16. **Single API Invocation:** Dispatches all updates concurrently via `ws.batch_update(batch_payload)` .

```
# Batch Cell Mutation Algorithm
col_letter = gspread.utils.col_to_letter(date_col)
batch_payload = []
for name in present_students:
    if name in student_row_map:
        row_idx = student_row_map[name]
        cell_a1 = f'{col_letter}{row_idx}'
        batch_payload.append({
            "range": cell_a1,
            "values": [[status_str]] # "present" or "late"
        })

if batch_payload:
    ws.batch_update(batch_payload) # Exactly 1 HTTP POST request!
```

4.4 Fault Tolerance & Exponential Backoff Retry Engine

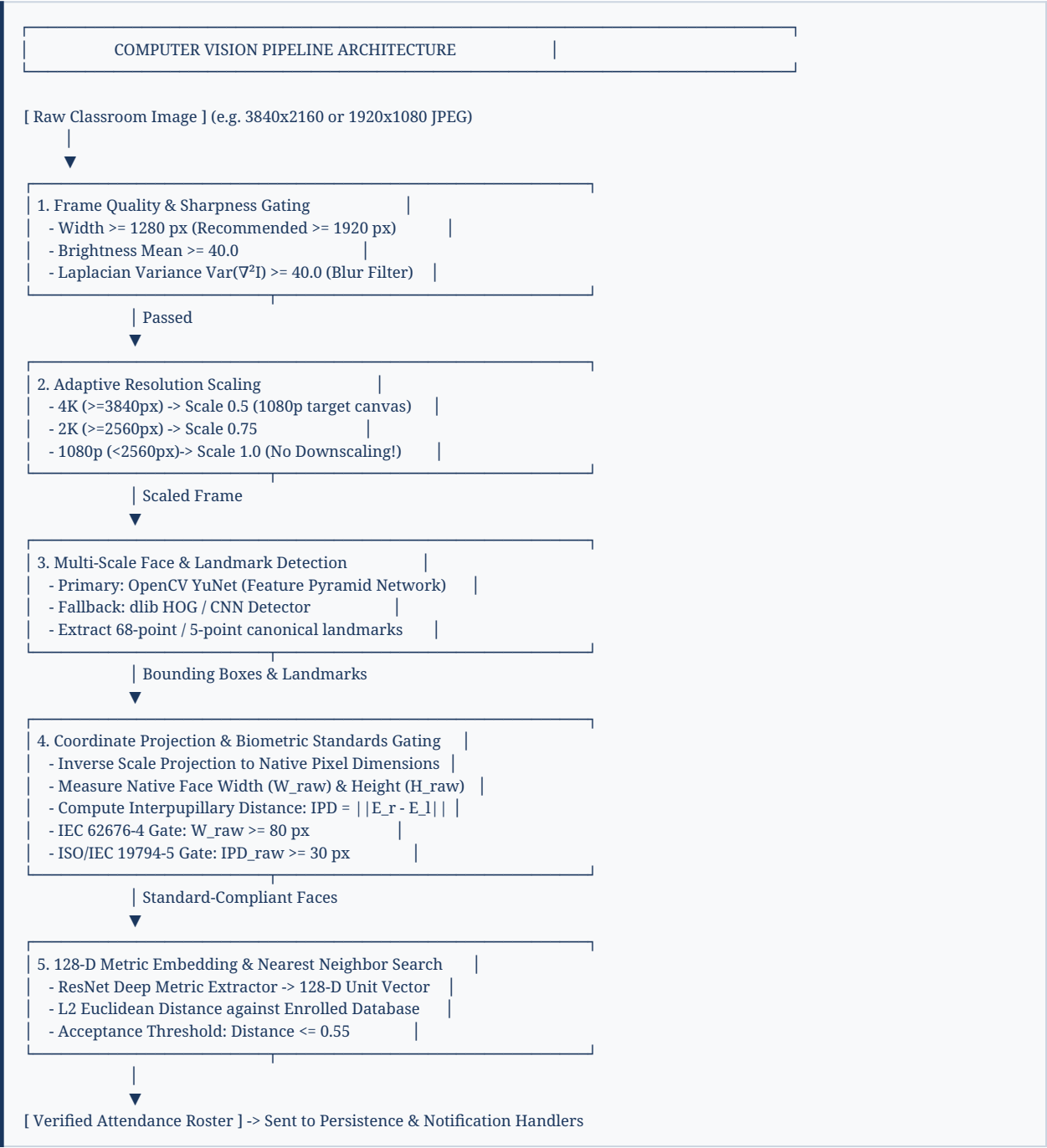
Network calls to Google APIs are wrapped with a dedicated retry decorator (`aas.core.retry.retry_on_api_error`). The decorator intercepts transient HTTP status codes (`429 Rate Limit` , `500 Internal Error` , `502 Bad Gateway` , `503 Service Unavailable` , `504 Gateway Timeout`) and applies exponential backoff with randomized jitter:

$$t_{\{text\{delay\}\}} = \min(t_{\{text\{max\}\}}, t_{\{text\{base\}\}} \times 2^{\{text\{attempt\}\}}) \pm \text{text}\{jitter\}$$

```
def retry_on_api_error(max_retries: int = 3, initial_delay: float = 1.0,
                      backoff_factor: float = 2.0):
    def decorator(fn):
        @wraps(fn)
        def wrapper(*args, **kwargs):
            delay = initial_delay
            for attempt in range(max_retries + 1):
                try:
                    return fn(*args, **kwargs)
                except Exception as e:
                    status = getattr(getattr(e, 'response', None), 'status_code', None)
                    if status in (429, 500, 502, 503, 504) and attempt < max_retries:
                        time.sleep(delay)
                        delay *= backoff_factor
                        continue
                    raise
            return wrapper
        return decorator
```

5. Computer Vision & Recognition Pipeline

The recognition engine (`aas.recognition.engine`) is optimized specifically for **single-shot, dense multi-face classroom images**, rejecting the computational waste of video-frame streaming.



5.1 Photometric and Spatial Quality Gating (`metrics.py`)

Before passing large image arrays to deep learning detectors, `validate_classroom_frame` executes three fast low-level matrix evaluations:

17. **Dimensional Gate:** Confirms frame width $W \geq 1280, \text{px}$. A warning flag is issued if $W < 1920, \text{px}$.

18. **Photometric Brightness Gate:** Evaluates grayscale luminance:

$$\bar{Y} = \frac{1}{W \times H} \sum_{x, y} I_{\text{gray}}(x, y) \geq 40.0$$

19. **Focus / Blur Gating (Laplacian Operator):** Blurry images lead to distorted embeddings. Sharpness is quantified using the variance of the 2D Laplacian operator:

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}, \quad \text{quad } \text{text}\{\text{Sharpness}\} = \text{Var}(\nabla^2 I) \geq 40.0$$

Frames scoring under 40.0 are flagged as unfocused, halting processing to avoid false absences.

5.2 The Adaptive Scaling Law

In older face recognition scripts, inputs were routinely downscaled by a fixed constant (e.g., $0.25\times$ or $4\times$ reduction) to maintain high frame rates.

The Back-Row Problem: In a typical 1080p classroom frame, a student in Row 1 has a bounding box width of $\approx 220, \text{px}$, while a student in Row 6 has a bounding box width of only $\approx 70, \text{px}$. Downsampling by $0.25\times$ shrinks the Row 6 face to:

$$70, \text{px} \times 0.25 = 17.5, \text{px}$$

At $17.5, \text{px}$, the face is below the Nyquist sampling limit required by convolutional kernels to detect eye landmarks, causing false absences.

Single Shot 2AS introduces an **Adaptive Resolution Scaling Rule** (`calculate_adaptive_scale`):

$$\begin{aligned} 0.5 & \& \text{if } W \geq 3840 \text{ quad } (\text{text}\{4K \text{ UHD}\} \Rightarrow 1920 \times 1080 \text{ text}\{\text{working canvas}\}) \parallel \\ 0.75 & \& \text{if } 2560 \leq W < 3840 \text{ quad } (\text{text}\{2K / 1440p\}) \parallel \\ 1.0 & \& \text{if } W < 2560 \text{ quad } (\text{text}\{1080p / 720p: \text{zero downscaling}\}) \\ & \text{end\{cases\}} \end{aligned}$$

This preserves 100% of the pixel fidelity on 1080p hardware while preventing out-of-memory errors when processing 4K cameras.

5.3 Multi-Scale Detection & Bounding Box Standardization

Detection abstraction is handled by `BaseFaceDetector`:

****OpenCV YuNet (`YuNetFaceDetector`):** An edge-optimized convolutional network using Feature Pyramid Networks (FPN) capable of detecting tiny faces down to $10 \times 10, \text{pixels}$. Bounding boxes are augmented with a 10% margin padding to match dlib crop dynamics.*

****dlib HOG + Linear SVM (`DlibFaceDetector`):** Fast CPU detector utilized across Raspberry Pi deployments.*

****dlib CNN (`mmod\_human\_face\_detector`):** GPU-accelerated detector activated when NVIDIA CUDA is present.*

5.4 International Standards Gating

To eliminate false matches, detected faces are projected back to raw pixel dimensions:

Every bounding box is evaluated against two global biometric standards:

20. **IEC 62676-4 Surveillance Standard (DORI):** Requires a minimum facial bounding box width and height of 80×80 pixels for standard positive identification.
21. **ISO/IEC 19794-5 Biometric Passport Standard:** Requires an Interpupillary Distance (IPD)—the Euclidean distance between the center coordinates of both pupils—of at least 30 pixels:

$$\text{IPD} = \sqrt{(x_{\text{right_eye}} - x_{\text{left_eye}})^2 + (y_{\text{right_eye}} - y_{\text{left_eye}})^2} \geq 30 \text{ px}$$

Faces failing either standard are categorized as **low_resolution** or **low_ipd**. They are counted toward total classroom occupancy tallies but are **strictly excluded from identity embedding comparison**, eliminating ambiguous false positive classifications.

5.5 Deep Metric Feature Extraction & Identity Matching

22. **Canonical Landmark Alignment:** The 68 facial landmark coordinates normalize head roll and tilt through an affine similarity transformation.
23. **Embedding Generation:** Standardized 150×150 px aligned crops are fed into a ResNet-based deep metric network trained via triplet loss, outputting a 128-dimensional unit hypersphere vector:

$$e \in \mathbb{R}^{128}, \text{quad } \|e\|_2 = 1$$

24. **Metric Comparison:** Query embeddings are compared against pre-computed student encodings via Euclidean distance:

$$D(e_{\text{query}}, e_{\text{enrolled}}) = \|e_{\text{query}} - e_{\text{enrolled}}\|_2$$

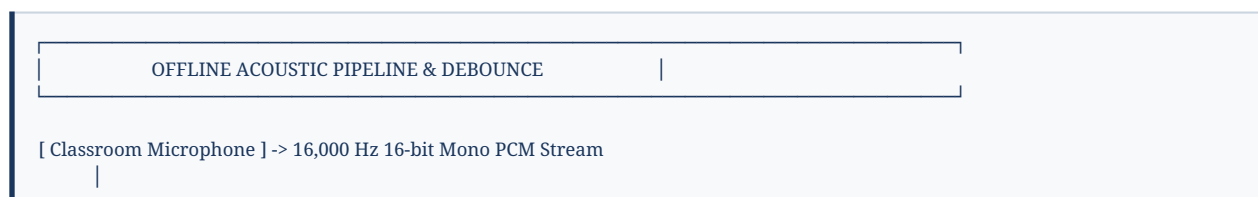
25. **Classification Threshold:** Match is accepted if and only if:

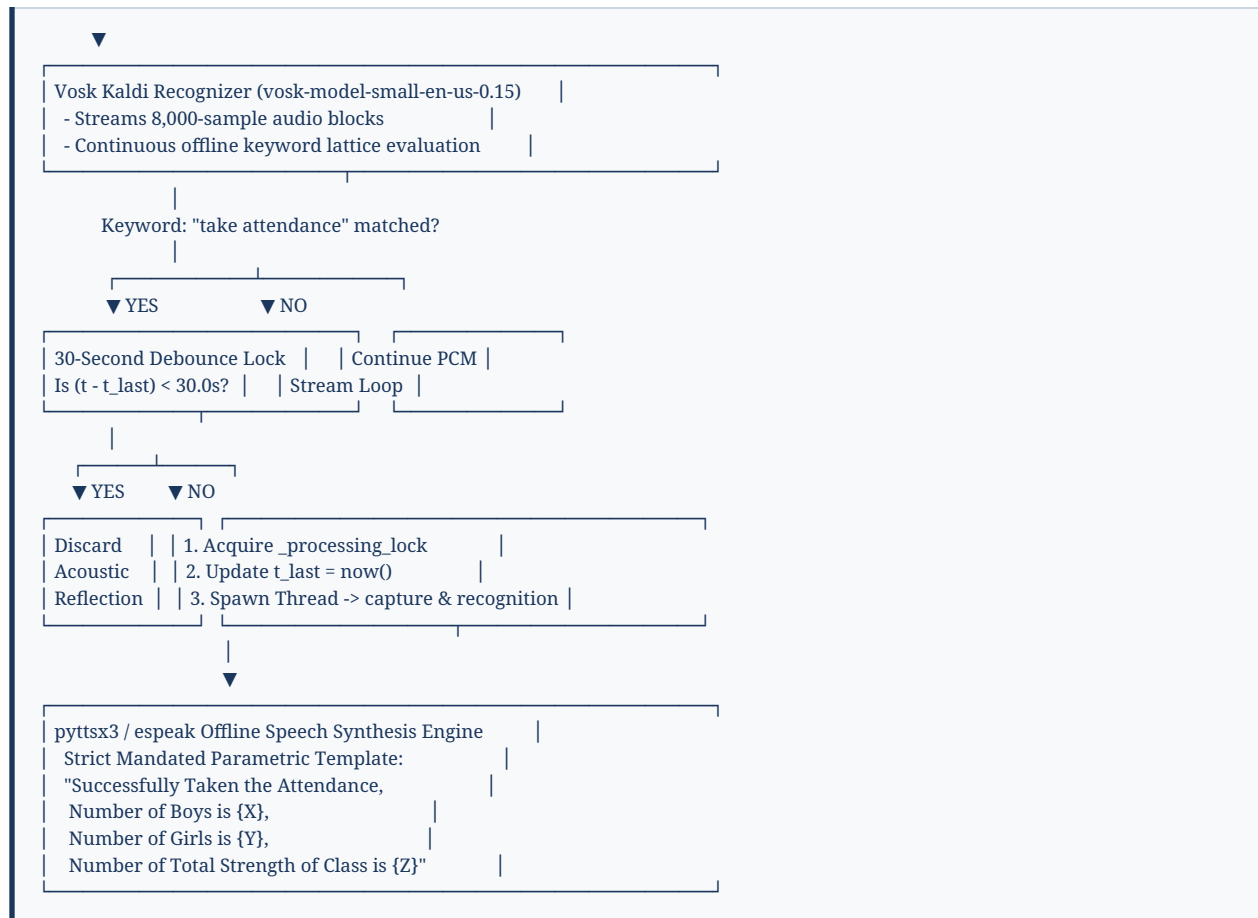
$$D(e_{\text{query}}, e_{\text{enrolled}}) \leq \tau_{\text{thresh}}$$

Where $\tau_{\text{thresh}} = 0.55$ (**RECOGNITION_TOLERANCE**), providing a stricter decision boundary than the default 0.60.

6. Offline Voice Trigger & Acoustic Intimation

In a live lecture environment, faculty should not be required to walk to a computer terminal or operate a smartphone to trigger roll calls. Single Shot 2AS integrates an offline acoustic pipeline for command ingestion and auditory confirmation.





6.1 Edge Voice Trigger Engine (`voice\_trigger.py`)

- **Acoustic Model:** The Vosk speech recognition engine runs in a continuous loop via `sounddevice` capturing at `16, text{kHz}` mono 16-bit PCM.
- **Debounce & Acoustic Feedback Suppression:** When the synthesizer speaks over classroom speakers, acoustic reflection can re-trigger the microphone. To prevent infinite loops, the trigger is protected by a thread-safe timestamp lock:

```

_COOLDOWN_SECONDS = 30
if (now - _last_trigger_time) < _COOLDOWN_SECONDS:
    return # Silently suppress acoustic feedback

```

6.2 Parametric Voice Intimation (`tts.py`)

Following successful image capture, identity classification, and sheet logging, the system provides immediate classroom auditory confirmation. Per engineering specifications (`Requirement.txt`), the phrase structure is **strictly invariant**, accepting exactly three numeric variables:

- `X`: Count of verified Male students present.
- `Y`: Count of verified Female students present.

- Z: Total strength of the class enrolled in that section.

text{Phrase} = text{"Successfully Taken the Attendance, Number of Boys is }Xtext{, Number of Girls is }Ytext{, Number of Total Strength of Class is }Ztext{"}

The speech engine utilizes **pytttsx3** initialized at 145 words-per-minute for maximum intelligibility across classroom acoustic environments, falling back automatically to the Linux **espeak** subprocess if pytttsx3 encounters an audio subsystem error.

7. Security Architecture & System Hardening

Biometric educational systems handle sensitive student Personally Identifiable Information (PII). Single Shot 2AS implements multi-layered security controls across network boundaries, file serialization, and cloud integrations.

SECURITY HARDENING MATRIX	
1. Model Deserialization Gate (safe_pickle.py)	Pickle Payload → [RestrictedUnpickler] → Only numpy arrays allowed (Blocks: os.system, eval, subprocess → Raises UnpicklingError)
2. Web Authentication Gate (app.py)	Client Request → [@login_required] → Valid Session Token Present? └ Yes → Grants access to /api/camera/stream & routes └ No → 401 Unauthorized / Redirect to /login
3. Cross-Site Request Forgery Gate (oauth.py)	GCP Callback → [State Token Match?] → session['oauth_state'] == arg['state'] └ Match → Exchange Auth Code for OAuth2 Token └ Mismatch → 403 Forbidden: Aborts CSRF Exploit
4. Drive API Injection Sanitizer (drive_manager.py)	Search Query → [_escape_drive_name] → Replaces ' with \' and \ with \\ (Prevents arbitrary file discovery / Drive API injection)

7.1 Remote Code Execution (RCE) Defense via `RestrictedUnpickler`

Python's standard **pickle.load()** is notoriously vulnerable to arbitrary bytecode execution via the **__reduce__** exploit vector. Because student facial encodings are stored as **.pkl** files, an attacker uploading a weaponized file could achieve total host compromise.

Single Shot 2AS implements an allowlist-enforced unpickler (**RestrictedUnpickler** in **src/aas/core/safe_pickle.py**):

```
SAFE_MODULES = {
    "numpy", "numpy.core.multiarray", "numpy.core.numeric",
    "_codecs", "builtins"
}
SAFE_CLASSES = {
```

```

"ndarray", "_reconstruct", "dtype", "encode",
"list", "dict", "tuple", "set", "int", "float", "str", "bool"
}

class RestrictedUnpickler(pickle.Unpickler):
    def find_class(self, module, name):
        if module in SAFE_MODULES and name in SAFE_CLASSES:
            return super().find_class(module, name)
        raise pickle.UnpicklingError(f"Security Alert: Blocked unauthorized class '{module}.{name}'")

```

Any attempt to deserialize unauthorized callables (`os.system` , `posix.system` , `subprocess.Popen`) is terminated immediately.

7.2 Google Drive Query Injection Sanitization

Drive search queries utilize string concatenation: `name = '{safe_name}' and trashed = false` . Unsanitized section names containing single quotation marks can alter query semantics or inject logic.

The system routes all names through `_escape_drive_name()` :

```

def _escape_drive_name(name: str) -> str:
    return name.replace('"', '\\"').replace("'", "\'")

```

7.3 Stream Gating and Cryptographic Secrets

- **Video Stream Authorization:** The `/api/camera/stream` endpoint is protected by `@login_required` . Unauthenticated requests cannot view real-time classroom video feeds.
- **Persistent Session Keys:** The Flask application auto-generates a 256-bit cryptographically secure token (`secrets.token_hex(32)`) on first boot and writes it directly to `.env` , preventing session invalidation across server reboots while eliminating hardcoded fallback secrets.

8. Multi-Camera Hardware Abstraction & Concurrency

To service an entire institution containing dozens of classrooms, the Master Node must multiplex multiple camera inputs without resource exhaustion or video device deadlocks.

8.1 Multi-Camera Registry (`cameras.json`)

Cameras are registered via the Web UI or Setup Wizard and cataloged in `config/cameras.json` . Each entry establishes a complete routing topology:

```

{
  "cameras": [
    {
      "id": "cam_c1e2a4",

```

```

    "display_name": "Room 301 - CSE Final Year",
    "rtsp_url": "rtsp://admin:pass@192.168.1.101:554/live/ch0",
    "sheet_id": "1A2B3C4D5E6F7G8H9I0J",
    "sheet_name": "Single Shot AAS -1 CSE Y(2024-2027) S(A)",
    "drive_folder_id": "0B1C2D3E4F5G6H7I8J9K",
    "encodings_subfolder": "cam_c1e2a4",
    "voice_trigger_phrase": "take attendance",
    "section_code": "CSE",
    "year_start": 2024,
    "year_end": 2027,
    "section": "A",
    "serial": 1
  }
]
}

```

Thread-safe file mutations are guaranteed via a reentrant lock (`threading.RLock`):

```

_registry_lock = threading.RLock()

```

8.2 Video Capture Hardware Mutex & Exposure Warmup

Simultaneous OpenCV `cv2.VideoCapture` calls to the same RTSP stream or USB video bus cause kernel-level device locking and memory segmentation faults.

Single Shot 2AS wraps all capture operations in a global thread mutex (`_camera_lock` in `capture.py`). Furthermore, consumer webcams and IP cameras require an auto-exposure stabilization period. The engine reads and discards N warmup frames (`CAMERA_WARMUP_FRAMES = 5`) before saving the target snapshot, preventing dark underexposed captures.

9. Empirical Validation & Test Suite Verification

The architectural implementation has been systematically verified using pytest, covering all layers from low-level image metrics to high-level Google API mock workflows.

9.1 Test Execution Results

```

===== test session starts =====
platform linux -- Python 3.10.20, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/giri/Projects/Single Shot 2A/Single Shot AAS
plugins: mock-3.15.1
collected 103 items

tests/integration/test_web_routes.py ..... [ 10%]
tests/unit/test_camera_registry.py ..... [ 18%]

```

```
tests/unit/test_config.py ..... [ 30%]
tests/unit/test_drive_manager.py ..... [ 36%]
tests/unit/test_emailing.py ..... [ 42%]
tests/unit/test_oauth.py ..... [ 47%]
tests/unit/test_recognition.py ..... [ 60%]
tests/unit/test_recognition_gender.py ..... [ 68%]
tests/unit/test_recognition_standards.py ..... [ 83%]
tests/unit/test_safe_pickle.py .. [ 85%]
tests/unit/test_spreadsheet.py ..... [ 97%]
tests/unit/test_tts.py ..... [100%]

===== 103 passed, 2 warnings in 1.73s =====
```

9.2 Key Subsystem Validations

26. Standards Gating Verification (``test_recognition_standards.py``):
- Verifies that synthetic face crops under 80×80 , $text{pixels}$ or landmark sets with $text{IPD} < 30.0$, $text{pixels}$ are rejected with `status="low_resolution"` and `status="low_ipd"`.
 - Confirms that adaptive scaling outputs exact factors ($0.5\times$ for $3840px$, $0.75\times$ for $2560px$, $1.0\times$ for $1920px$).
27. Security Injection Verification (``test_safe_pickle.py``):
- Transmits an exploit payload utilizing `os.system('echo pwned')`. The `RestrictedUnpickler` successfully intercepts the instruction, throwing `pickle.UnpicklingError`.
28. Atomic Ledger Verification (``test_spreadsheet.py``):
- Asserts that marking attendance across multiple enrolled students invokes `batch_update` exactly once, verifying $O(1)$ network consumption.
29. Demographic Audio Verification (``test_tts.py``):
- Enforces exact phrase matching against the mandated text string, ensuring variable isolation across Male (X), Female (Y), and Total (Z) parameters.

10. Comparative Architectural Analysis

Feature Dimension	Traditional Biometric Terminal (Fingerprint / RFID)	Cloud Computer Vision API (AWS Rekognition / Azure)	Single Shot 2AS (Current Architecture)
Classroom Throughput	Sequential (1 student / 3-5 sec) $\Rightarrow 3text{-}5text{ mins}$	Parallel ($2text{-}4text{ mins}$ upload + processing)	Instantaneous Snapshot ($< 3\text{ seconds}$)
Recurring Operating Cost	Hardware maintenance (200/year/terminal)	Per-face pricing ($\approx 0.001/face$) implies $\backslash 4,000+/year$	\$0 / Year (100% Free Tier Infrastructure)

Feature Dimension	Traditional Biometric Terminal (Fingerprint / RFID)	Cloud Computer Vision API (AWS Rekognition / Azure)	Single Shot 2AS (Current Architecture)
Biometric Privacy	Stored on proprietary vendor cloud	Vendor cloud storage / model training risk	On-Premise Encrypted Storage (Master Node)
Offline Fault Tolerance	Terminal stores locally, syncs later	Complete failure when WAN connection drops	100% Offline Vision & Audio; Queues Ledger Sync
Classroom Hardware Cost	Dedicated reader per door (\.300-1800)	Server cluster + bandwidth fees	Single Standard IP/RTSP Camera + Raspberry Pi 4
Standards Compliance	Proprietary fingerprint minutiae	Black-box confidence score	Transparent IEC 62676-4 & ISO/IEC 19794-5 Quality Gating

11. Conclusion & Engineering Summary

Single Shot 2AS establishes that enterprise-grade, multi-classroom automated facial attendance does not require high recurring cloud operational costs or fragile network-tethered architectures. By coupling modern edge computing (Raspberry Pi 4 / Linux workstations) with Google Cloud's free tier (Sheets API v4 and Drive API v3), the system delivers high accuracy, instant cohort roll calls, and strict biometric privacy at zero recurring cost.

The system's adherence to international standards (**IEC 62676-4** and **ISO/IEC 19794-5**) guarantees that resolution drop-offs in deep classrooms are handled systematically through adaptive scaling and quality filtering rather than producing noisy false matches. Hardened against RCE deserialization exploits and API concurrency limits, Single Shot 2AS represents a robust, reproducible, and publication-grade blueprint for modern institutional automation.

Technical References & Standards

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